VINEYARD
STAY CONNECTED

WASTEWATER COLLECTION SYSTEM IMPACT FEE FACILITY PLAN

(HAL Project No.: 319.10.100)

Adopted May 28, 2025

HANSEN ALLEN & LUCE inc
ENGINEERS

VINEYARD CITY

WASTEWATER COLLECTION AND TREATMENT SYSTEM IMPACT FEE FACILITY PLAN

(HAL Project No.: 319.10.100)



Ridley J. Griggs, P.E.

Project Manager

HANSEN ALLEN & LUCE INC
ENGINEERS

Adopted May 28, 2025

IMPACT FEE CERTIFICATION

The Utah Impact Fee Act requires certification for the Impact Fee Facilities Plan (IFFP). Hansen, Allen & Luce provides this certification with the understanding that the recommendations in the IFFP are followed by City Staff and elected officials. If all or a portion of the IFFP is modified or amended, or if assumptions presented in this analysis change substantially, this certification is no longer valid. All information provided to Hansen, Allen & Luce, Inc. is assumed to be correct, complete, and accurate.

IFFP Certification

Hansen, Allen & Luce, Inc. certifies that the Impact Fee Facilities Plan (IFFP) prepared for the wastewater collection system:

1. includes only the costs of public facilities that are:
    a. allowed under the Impact Fees Act; and
    b. actually incurred; or
    c. projected to be incurred or encumbered within six years after the day on which each impact fee is paid;
2. does not include:
    a. costs of operation and maintenance of public facilities;
    b. costs for qualifying public facilities that will raise the level of service for the facilities, through impact fees, above the level of service that is supported by existing residents;
    c. an expense for overhead, unless the expense is calculated pursuant to a methodology that is consistent with generally accepted cost accounting practices and the methodological standards set forth by the federal Office of Management and Budget for federal grant reimbursement; and
3. complies in each and every relevant respect with the Impact Fees Act.

HANSEN, ALLEN & LUCE, INC.

TABLE OF CONTENTS

Page No

IMPACT FEE SUMMARY……..……...……iii

CHAPTER 1 – INTRODUCTION

1.1 Background 1-1
1.2 Purpose....1-1
1.3 Impact Fee Collection....1-1
1.4 Master Planning 1-2

CHAPTER 2 – EXISTING AND FUTURE WASTEWATER COLLECTION SYSTEM

2.1 General 2-1
2.2 Existing and Future Equivalent Residential Units 2-1
2.3 Level of Service....2-3
2.4 Methodology Used to Determine Existing System Capacity 2-3
2.5 Collections....2-3
2.6 Capital Facilities to Meet System Deficiencies 2-3

CHAPTER 3 – IMPACT FEE FACILITIES PLAN

3.1 General 3-1
3.2 Growth Projections 3-1
3.3 Capacity of Existing Facilities 3-2
3.4 Capacity of Future Facilities 3-2

REFERENCES R-1

Appendix A
Information from the Vineyard Wastewater Collection System Master Plan

Appendix B
Estimated Future Project Costs

LIST OF FIGURES

Page No

Figure 3-1: Areas of Projected Growth ..... After 3-1

LIST OF TABLES

Page No

Table 2-1: System ERUs 2-1
Table 2-2: TSSD System I&I Projections 2-2
Table 2-3: Orem System I&I Projections 2-2
Table 2-4: Existing Pipe Capacity Deficiencies and Estimated Cost 2-3
Table 3-1: Growth Projections 3-1
Table 3-2: Impact Fee Eligibility of Existing Collection Facilities 3-2
Table 3-3: Estimated Cost of Future Collection Facilities 3-3
Table 3-4: System Flow Projections to TSSD 3-3
Table 3-5: System Flow Projections to Orem 3-4
Table 3-6: Planning Component of Impact Fee 3-4

IMPACT FEE SUMMARY

In recent years, construction costs have risen substantially due to a number of factors, including material shortages, labor shortages, and supply chain constraints. To properly account for these changes, Vineyard City commissioned this impact fee update.

This Impact Fee Facility Plan addresses the **collection system**. Vineyard's existing collection system conveys wastewater to the TSSD system and Orem WRF, where it is treated. The impact fee **service area** is outlined in Figures 2-1 and 2-2 of the Wastewater Collection System Master Plan, which are included in Appendix A for reference.

The impact fee unit for wastewater collection is based on the Equivalent Residential Unit (ERU). An ERU is equal to the average hydraulic loading of one residential connection. The method of using ERUs for analysis is a way for allocating existing and future demands of non-residential land uses. The level of service for wastewater collection is 160 gpd per ERU.

The existing system served about 5,878 ERUs at the end of 2023. Projected **growth** adds 10,598 equivalent residential units in the next 10 years for a total of 16,476 connections or equivalent by 2033.

Impact fee calculations are based on the proportional historical buy-in costs of **excess capacity** in existing facilities and **new projects** required entirely to provide capacity for new development. The cost of providing capacity to resolve existing deficiencies is not included in the impact fee. Available capacity in existing facilities and capacity that is created through new projects is included in the impact fee. In addition to the proportionate share of costs of existing facilities, the impact fee is based on infrastructure that will be constructed within the next 10 years.

The purpose of this Impact Fee Facilities Plan is to provide key information to support the calculation of an impact fee. Impact fee calculations will be included in a separate Impact Fee Analysis document.

CHAPTER 1
INTRODUCTION

1.1 Background

Vineyard is located in northern Utah County. Vineyard owns and manages a wastewater collection system that serves the City. Most wastewater from Vineyard is conveyed to the water reclamation facility operated by Timpanogos Special Services District (TSSD). A small portion of the wastewater generated in the City drains south to the Orem City collection system and is subsequently treated at the Orem Water Reclamation Facility (Orem WRF).

1.2 Purpose

The City has recognized the need to plan for increased demands on its wastewater collection system as a result of growth. To do so, an Impact Fee Facility Plan (IFFP) was prepared to form the basis for an Impact Fee Analysis (IFA), which will allow the City to charge an impact fee to help pay for capital projects necessary to support future growth.

In recent years, construction costs have risen substantially due to a number of factors, including material shortages, labor shortages, and supply chain constraints. To account for rising construction costs, Vineyard City commissioned this impact fee update.

This report identifies those items that the Utah Impact Fees Act specifically requires, including demands placed upon existing facilities by new development and the proposed means by which the municipality will meet those demands. The Wastewater Collection Master Plan that was prepared in 2024 was also used to support this analysis. The master plan identified several growth-related projects needed within the 10-year planning window. Therefore, the calculated impact fee is based on excess capacity and documented historic costs, as well as future capital projects.

1.3 Impact Fee Collection

Impact fees enable local governments to finance public facility improvements necessary for growth, without burdening existing customers with costs that are exclusively attributable to growth.

An impact fee is a one-time charge on new development to pay for that portion of a public facility that is required to support that new development.

In order to determine the appropriate impact fee, the cost of the facilities associated with future development must be proportionately distributed. As a guideline in determining the “proportionate share,” the fee must be found to be roughly proportionate and reasonably related to the impact caused by the new development.

1.4 Master Planning

The Vineyard City Wastewater Collection System Master Plan was prepared in 2024 and is incorporated by reference into this analysis. The master plan for the City's wastewater collection system is more comprehensive than the IFFP. It provides the basis for the IFFP and identifies all capital facilities required for the wastewater system inside the 20-year planning range, including maintenance, repair, replacement, and growth-related projects.

The recommendations made within the master plan are in compliance with current City policies and standard engineering practices.

A hydraulic model of the wastewater collection system was used to complete the Wastewater Collection System Master Plan. The model was used to assess existing performance, to establish a proposed level of service, and to confirm the effectiveness of the proposed capital facility projects to maintain the proposed level of service over the next 10 years.

CHAPTER 2
EXISTING AND FUTURE WASTEWATER COLLECTION SYSTEM

2.1 General

The purpose of this chapter is to provide information regarding the existing wastewater collection system, identify the current and proposed levels of service, and analyze the remaining capacity of the existing system's facilities.

Vineyard's existing wastewater collection system is comprised of gravity pipes including laterals, collectors, interceptors, lift stations, and force mains. Most wastewater generated within the service area is conveyed to an outfall to the TSSD system, although a small portion of the generated wastewater is conveyed to Orem WRF. Figures 2-1 and 2-2 of the Wastewater Collection System Master Plan illustrate the existing wastewater system and are included for reference in Appendix A.

2.2 Existing and Future Equivalent Residential Units

In order to compare the relative quantities of wastewater loading between different types of land use, it is helpful to use a common unit of measure. The unit of measure that is used with this analysis is the Equivalent Residential Unit (ERU). The use of ERUs is a typical approach to describe the wastewater collection system's usage. An ERU is equal to the average loading of residential connections. Once the ERU is established, non-residential uses can be quantified in terms of multiples or fractions of an ERU. An ERU is the ratio of non-residential wastewater loadings in comparison to an equivalent residential level of service. For this analysis all residential connections, including townhouses and apartments were equated to one ERU for indoor water demands.

Table 2-1 is a summary of ERUs in the existing system and the system at the 10-year design condition as described in the Vineyard City Wastewater Collection System Master Plan.

Table 2-1
System ERUs

Condition	Treated at TSSD (ERUs)	Treated at Orem WRF (ERUs)	Total ERUs
Existing	5,592	286	5,878
10-Year	15,996	479	16,476
Difference	**+10,404**	**+193**	**+10,598**

2.3 Level of Service

The level of service designated for the wastewater collection system has been established by the City to provide adequate wastewater collection capacity. It is based on analysis of data as described in the Wastewater Collection System Master Plan.

ERU Loading and Treatment

- The existing level of service is for average daily flow is 160 gpd per ERU.

Infiltration and Inflow

Infiltration and inflow (I&I) occur in wastewater collection systems. While Vineyard City has employed and will continue to employ a number of techniques to manage I&I, it is not practical or cost-effective to completely eliminate it. During the master plan, pipes were sized to accommodate expected quantities of I&I for existing and future scenarios. Tables 2-2 and 2-3 summarize the I&I projections as described in the Wastewater Collection System Master Plan.

Table 2-2
TSSD System I&I Projections

Condition	Inflow (MGD)	Infiltration (MGD)
Existing	1.15	0.12
10-Year	1.28	0.15
Buildout	1.5	0.20

Table 2-3
Orem System I&I Projections

Condition	Inflow (GPD)	Infiltration (GPD)
Existing	15,000	2,000
10-Year	25,100	3,350
Buildout	25,000	3,350

These flow rates are included in the hydraulic model.

Wastewater Collection System Capacity

- Peak daily flow in the pipe must not exceed a depth/diameter ratio of 0.75 for pipe diameters 12 inches and greater and 0.50 for pipe diameters less than 12 inches. The remaining capacity is reserved for unexpected flows, peaking, or flow restrictions.
- Per State of Utah standards, no newly installed collection pipe may be less than 8 inches in diameter.

2.4 Methodology Used to Determine Existing System Capacity

The method for determining the remaining capacity in the wastewater collection system was based on the defined level of service in terms of ERUs. The difference between the ERU capacity and ERU existing demand for each component is the remaining capacity.

Collection System Capacity

A hydraulic model was developed for the purpose of assessing system operation and capacity taking into account additional water that results from I&I. Inflow for Vineyard was assessed by evaluating wastewater flow records before and after storm events. Infiltration was determined by comparing billed winter water sales to wastewater flow records. For the collection system, the model was used to calculate remaining system capacity in terms of ERUs.

Wastewater Treatment Capacity

Vineyard City maintains agreements with Orem City and TSSD for wastewater treatment. Treatment capacity is as specified in the City's existing agreements.

2.5 Collections

The existing Vineyard City wastewater collection system consists of nearly 36 miles of pipeline and approximately 922 manholes. The pipes range in size from 8-inch diameter to 36-inch diameter. Lift stations are used to pump wastewater where gravity flow sewers are not capable of conveying flow to the TSSD outfall.

2.6 Capital Facilities to Meet System Deficiencies

The existing wastewater collection system is generally adequate to convey the anticipated wastewater to the wastewater treatment plant. However, there are a few areas with inadequate capacity, which are described in Table 2-4 below (same as Table 6-4 in the Wastewater Collection System Master Plan). Estimated costs for improving pipe deficiencies were also included in the table (same as Table ES-4 in Wastewater Collection System Master Plan). These projects are not eligible for impact fee reimbursement and are not discussed further in this report.

Table 2-4
Existing Pipe Capacity Deficiencies and Estimated Cost

Project ID	Location	Cost
E-1	Mallard Drive between 400 North and 800 North	$1,147,000
E-2	Next to railroad between 400 North and 800 North	$1,345,000

CHAPTER 3
IMPACT FEE FACILITIES PLAN

3.1 General

This chapter relies on the data presented in the previous chapters to determine impact fee eligibility for projects planned in the next 10 years to increase capacity for new growth and an appropriate buy-in cost of available existing excess capacity previously purchased by the City.

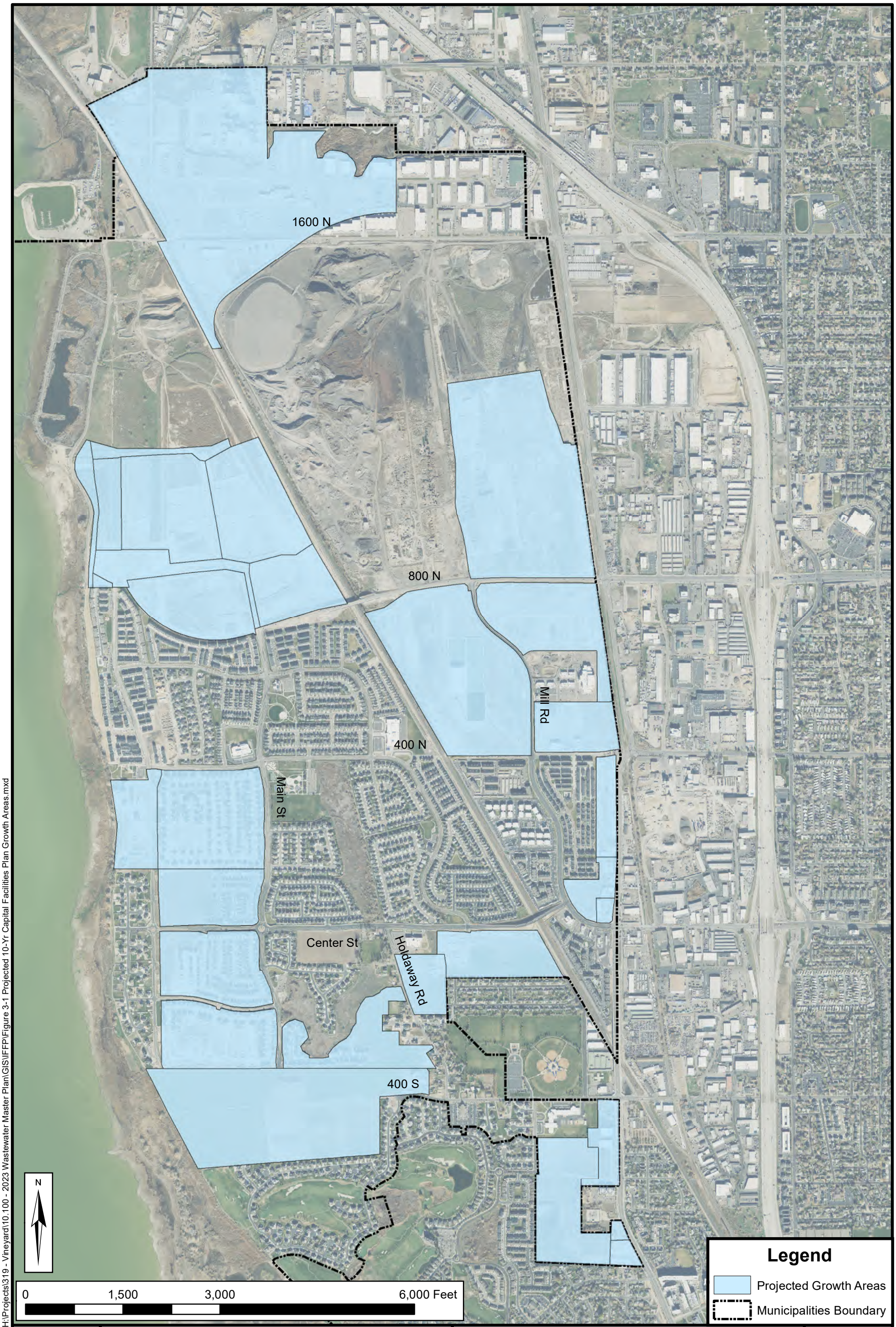
The wastewater collection system facility projects planned in the next 10 years to increase capacity for new growth included within the IFFP are presented.

3.2 Growth Projections

The development of impact fees requires growth projections over the next ten years. Growth projections for Vineyard were made using the rates provided in the Wastewater Collection System Master Plan. The existing system serves about 5,878 ERUs. Projected growth adds 10,598 ERUs in the next 10 years for a total of 16,476 ERUs. Total growth projections are summarized in Table 3-1. Further information on growth projections can be found in Figure 5-1 of the Wastewater Collection System Master Plan. This has been included in Appendix A for reference. The projected 10-year growth is shown in Figure 3-1.

Table 3-1
Growth Projections

Year	Total ERUs
2023	5,878
2024	6,358
2025	7,462
2026	9,141
2027	10,456
2028	11,549
2029	12,598
2030	13,646
2031	14,586
2032	15,536
2033	16,476
Change	**+10,598**



Date: 6/24/2024
Document Path: H:\Projects\319 - Vineyard\10.100 - 2023 Wastewater Master Plan\GIS\IFFP\Figure 3-1 Projected 10-Yr Capital Facilities Plan Growth Areas.mxd

HANSEN ALLEN & LUCE inc
ENGINEERS

VINEYARD CITY WASTEWATER SYSTEM

PROJECTED 10-YEAR GROWTH AREAS

FIGURE
3-1

3.3 Capacity of Existing Facilities

Future growth can be served either by excess capacity in existing facilities or by constructing new facilities. This section will focus on the existing capacity within the existing facilities.

Collection

The impact fee eligibility for the existing collection facilities is shown in Table 3-2. Capacity remaining in existing facilities was conservatively estimated as the difference between the existing ERU count and the projected ERU count at buildout. The cost of each facility associated with its remaining capacity is attributable to growth and can be counted towards the impact fee.

Table 3-2
Impact Fee Eligibility of Existing Collection Facilities

Existing ERU Count	Projected Buildout ERU Count	Difference	% Impact Fee Eligible
5,878	29,122	23,244	80%

Treatment

Vineyard City has secured treatment capacity in TSSD and Orem WRF.

3.4 Capacity of Future Facilities

This section will discuss the impact fee eligible costs for future facilities for the wastewater collection system.

Collection

The facilities and costs presented in Table 3-3 are proposed collection projects essential to maintain the current level of service while accommodating future growth within the next 10 years. The facility sizing for the future proposed projects were developed in the 2024 Wastewater Collection System Master Plan. They were based on the level of service, the City's land use plan, and hydraulic modeling. All future projects have a design life greater than 10 years, as required by the Impact Fee Act. Detailed cost estimates are included in Appendix B. Depictions of these projects are shown in Figure 6-4 of the Wastewater Collection System Master Plan, which has been included in Appendix A for reference. Further details are included in the Wastewater Collection System Master Plan.

Table 3-3
Estimated Cost of Future Collection Facilities

Project	Map ID[1]	Total Cost
Install 1,700 ft of 10” gravity line	10-1	$858,000
Install 2,400 ft of 15” gravity line	10-2	$1,337,000
Install 1,200 ft of 12” gravity line and 4,600 ft of 15” gravity line	10-3	$3,193,000
Install 4,500 ft of 8” gravity line	10-4	$2,192,000
Install 1,800 ft of 8” gravity line, 1,700 ft of 10” gravity line, and 3,050 ft of 15” gravity line	10-5	$3,434,000
Install two new pumps at Lift Station 2	10-LS2	$192,000
Total		**$11,206,000**

1. Refer to Figure 6-4 of the Wastewater Collection System Master Plan for the project number. This figure is included in Appendix A for reference.

Treatment

TSSD and Orem WRF currently manage the treatment facilities and all upgrades. Table 3-4 shows the existing and projected monthly average wastewater generated in the areas treated by TSSD.

Table 3-4
System Flow Projections to TSSD

Approximate Year	Total ERUs	Customer Flow Generation (MGD)	Inflow (MGD)	Infiltration (MGD)	Peak Daily Flow (MGD)	Average Monthly Flow (MGD)
2023	5,592	0.89	1.15	0.12	2.16	1.01
2033	15,996	2.56	1.28	0.15	3.99	2.71

Table 3-5 shows the existing and projected monthly average wastewater generated in the areas treated by the Orem WRF.

Table 3-5
System Flow Projections to Orem

Approximate Year	Total ERUs	Customer Flow Generation (GPD)	Inflow (GPD)	Infiltration	Peak Daily Flow (GPD)	Average Monthly Flow (GPD)
2023	286	45,760	15,000	2,000	62,760	47,760
2033	479	76,640	25,100	3,350	105,090	79,990

Planning

Planning services are also needed to support growth. The City intends to update their master plans approximately every 5 years and their impact fee studies approximately every 3 years. Considering this schedule, and the cost of the most recent impact fee updates, the ERUs served by each plan update were calculated as shown in Table 3-9.

Table 3-6
Planning Component of Impact Fee

Planning Document	Cost	% of Plan Associated with Growth[1]	ERUs Served[2]
2024 Wastewater Collection System Master Plan	$76,400.00	60%	5,671
2024 IFFP and IFA	$17,800.00	100%	3,263
Total	**$94,200.00**	**-**	**-**

1. Percentages to growth for the master plan was based on a review of the scope of the plan and associated fees for tasks associated with the existing system and future growth. The IFFP and IFA are 100% associated with growth.
2. ERUs served was defined as the amount of ERUs expected to develop during the 5-year life of the master plan and the 3-year life of the IFFP and IFA, respectively. See Table 3-1.

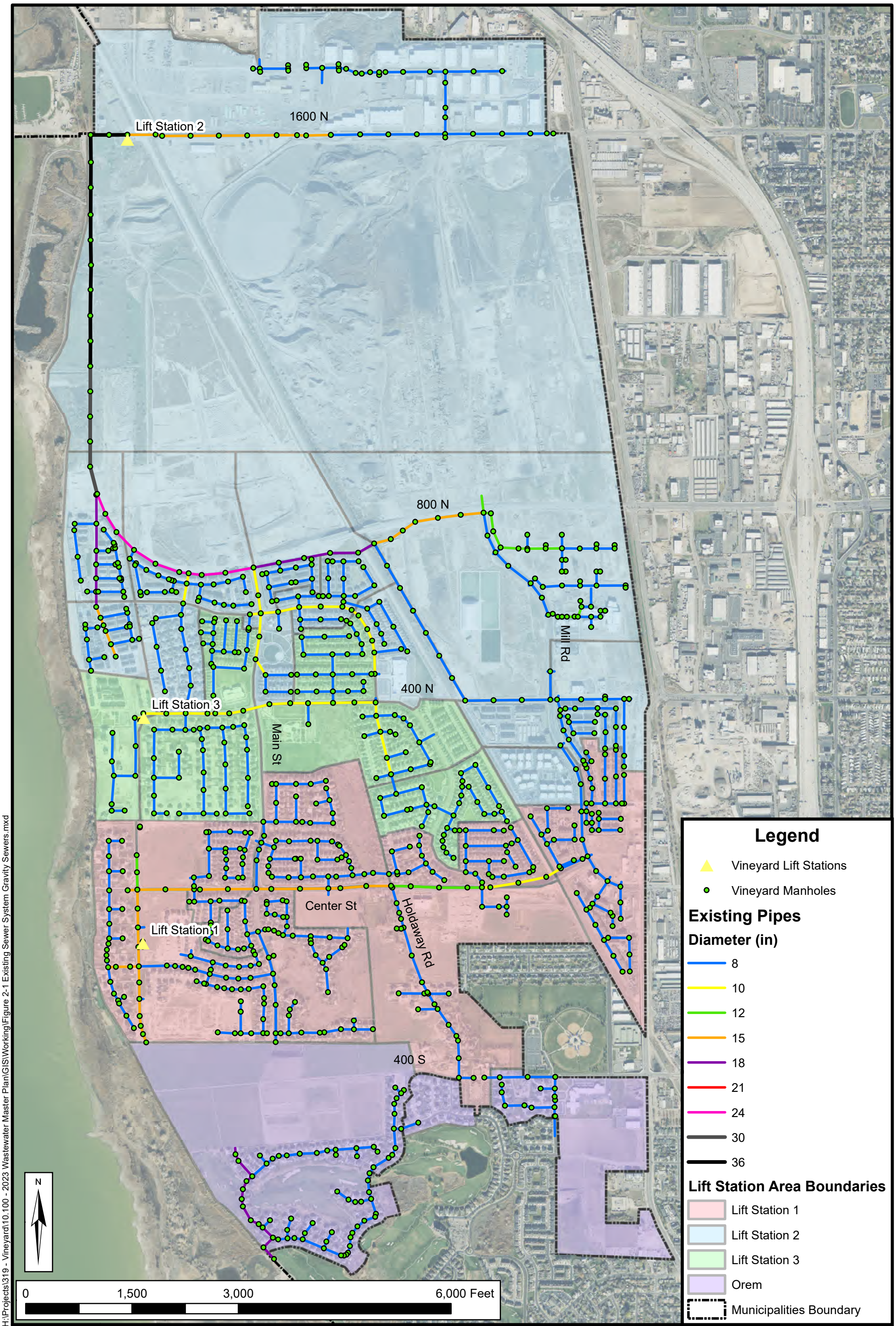
REFERENCES

State of Utah. 2014c. Utah Code Annotated, Section Utah Code 11-36a: Impact Fees Act

Hansen, Allen & Luce. 2024. “Vineyard Wastewater Collection System Master Plan”

APPENDIX A

Information from the Vineyard City Wastewater Collection System Master Plan



Date: 4/18/2024
Document Path: H:\Projects\319 - Vineyard\10.100 - 2023 Wastewater Master Plan\GIS\Working\Figure 2-1 Existing Sewer System Gravity Sewers.mxd

HANSEN ALLEN & LUCE INC ENGINEERS

VINEYARD CITY WASTEWATER SYSTEM

EXISTING SEWER SYSTEM GRAVITY SEWERS

FIGURE 2-1

Goes to TSSD outfall

1600 N

Lift Station 2

800 N

Mill Rd

400 N

Lift Station 3

Main St

Center St

Holdaway Rd

Lift Station 1

400 S

N

0 1,500 3,000 6,000 Feet

Legend

Vineyard Lift Stations

Force Mains

Diameter (in)

8

14

Existing Pipes

Diameter (in)

8

10

12

15

18

21

24

30

36

Lift Station Area Boundaries

Lift Station 1

Lift Station 2

Lift Station 3

Orem

Municipalities Boundary

Date: 6/4/2024
Document Path: H:\Projects\319 - Vineyard\10.100 - 2023 Wastewater Master Plan\GIS\Working\Figure 2-2 Existing Sewer System Force Mains.mxd

HANSEN ALLEN & LUCE INC ENGINEERS

VINEYARD CITY WASTEWATER SYSTEM

EXISTING SEWER SYSTEM FORCE MAINS

FIGURE 2-2

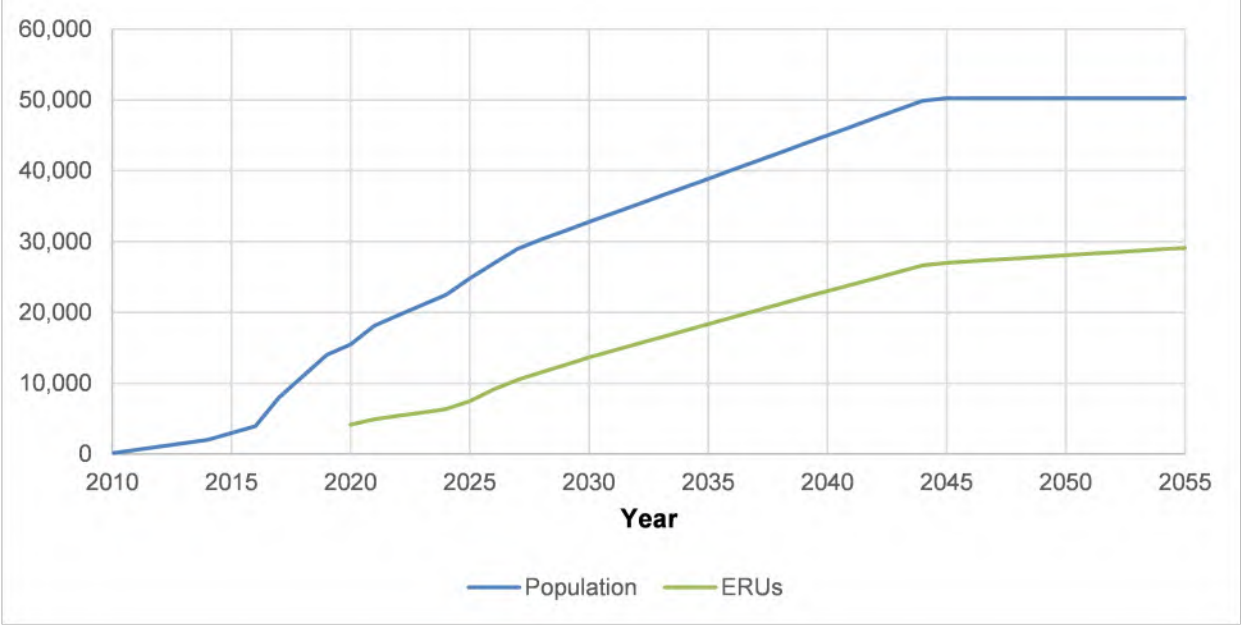
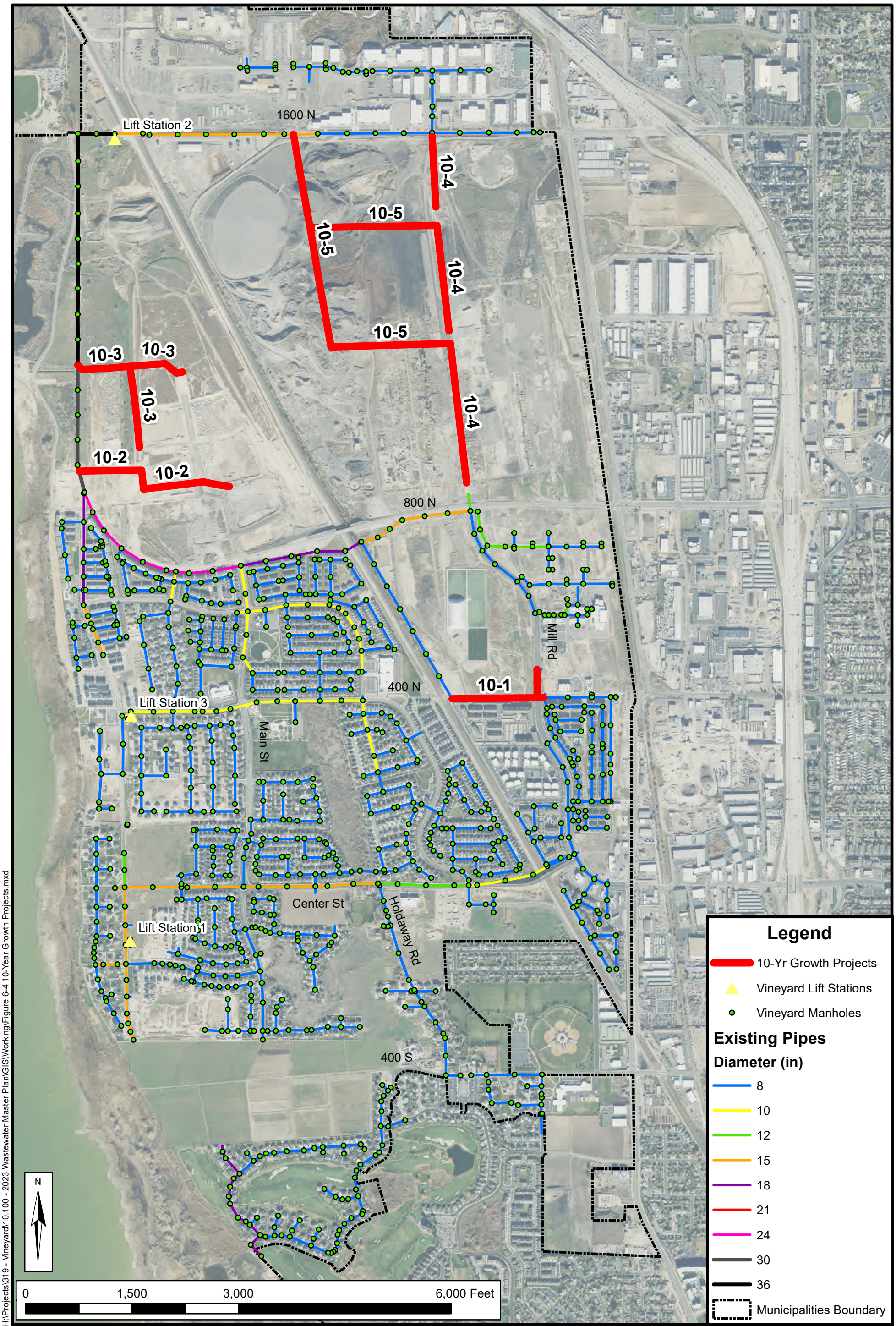


FIGURE 5-1 POPULATION AND ERU PROJECTIONS



Date: 4/30/2024
Document Path: H:\Projects\319 - Vineyard\10.100 - 2023 Wastewater Master Plan\GIS\Working\Figure 6-4 10-Year Growth Projects.mxd

HANSEN ALLEN & LUCE inc ENGINEERS

VINEYARD CITY WASTEWATER SYSTEM

10-YEAR GROWTH PROJECTS

FIGURE 6-4

APPENDIX B

Estimated Future Project Costs

Vineyard City Capital Facility Plan
Wastewater 10-Year Recommended Improvements
Preliminary Engineers Cost Estimates

	Item	Unit	Unit Price	Quantity	Total Price
10-1.	***400 N in front of school Sewer Line***				
	Install 10" gravity line	LF	$ 421	1700	$ 714,888
				Total	$ 714,888
				Engineering & Admin. (10%)	$ 71,489
				Contingency (10%)	$ 71,489
				Total to 400 N in front of school Sewer Line	**$ 858,000**
10-2.	***Future Downtown Area Sewer Lines***				
	Install 15" gravity line	LF	$ 464	2400	$ 1,114,087
				Total	$ 1,114,087
				Engineering & Admin. (10%)	$ 111,409
				Contingency (10%)	$ 111,409
				Total to Future Downtown Area Sewer Lines	**$ 1,337,000**
10-3.	***Future Downtown Area Sewer Lines***				
	Install 12" gravity line	LF	$ 438	1200	$ 525,301
	Install 15" gravity line	LF	$ 464	4600	$ 2,135,333
				Total	$ 2,660,635
				Engineering & Admin. (10%)	$ 266,063
				Contingency (10%)	$ 266,063
				Total to Future Downtown Area Sewer Lines	**$ 3,193,000**
10-4.	***Future Mill Road Sewer Lines***				
	Install 8" gravity line	LF	$ 406	4500	$ 1,826,400
				Total	$ 1,826,400
				Engineering & Admin. (10%)	$ 182,640
				Contingency (10%)	$ 182,640
				Total to Future Mill Road Sewer Lines	**$ 2,192,000**
10-5.	***Future East Geneva Area***				
	Install 8" gravity line	LF	$ 406	1800	$ 730,560
	Install 10" gravity line	LF	$ 421	1700	$ 714,888
	Install 15" gravity line	LF	$ 464	3050	$ 1,415,819
				Total	$ 2,861,267
				Engineering & Admin. (10%)	$ 286,127
				Contingency (10%)	$ 286,127
				Total to Future East Geneva Area	**$ 3,434,000**
10-LS2	***Lift Station 2 Upgrades***				
	Lift Station 2 - New Pumps and Motors	LS	$ 160,000	1	$ 160,000
				Total	$ 160,000
				Engineering & Admin. (10%)	$ 16,000
				Contingency (10%)	$ 16,000
				Total to Lift Station 2 Upgrades	**$ 192,000**

Total Costs $ 11,206,000